

Smart Bus Garage System - Comprehensive API Reference (v3.0)

This document is automatically generated from the OpenAPI specification and serves as the single source of truth for the system's REST API and WebSockets.

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Endpoints

Register a new user account

POST `/api/v1/auth/signup`

Create a new user account.

- Checks that the email is not already registered.
- Hashes the password with bcrypt before storing.
- Returns the created user (without password).

Request Body

```
{
  "email" (Required): string,
  "password" (Required): string,
  "full_name" (Required): string,
  "role": string,
  "phone": any,
}
```

Responses

Status **201** : Successful Response

```
{
  "id" (Required): integer,
  "email" (Required): string,
  "full_name" (Required): string,
  "role" (Required): string,
  "phone": any,
  "is_active" (Required): boolean,
  "created_at" (Required): string,
}
```

Status 422 : Validation Error

```
{
  "detail": [ {
    "loc" (Required): [ any ],
    "msg" (Required): string,
    "type" (Required): string,
    "input": any,
    "ctx": object,
  } ],
}
```

Login with email and password

POST /api/v1/auth/login

Authenticate a user with email (as username) and password.

- Uses OAuth2 password flow (form data: username + password).
- Returns a JWT access token on success.
- The 'username' field should contain the user's email address.

Request Body

None

Responses

Status 200 : Successful Response

```
{
  "access_token" (Required): string,
  "token_type": string,
}
```

Status 422 : Validation Error

```
{
  "detail": [ {
    "loc" (Required): [ any ],
    "msg" (Required): string,
    "type" (Required): string,
    "input": any,
    "ctx": object,
  } ],
}
```

Get current authenticated user

GET /api/v1/auth/me

Return the currently authenticated user's profile.

Requires a valid JWT token in the Authorization header: Authorization: Bearer

Authorization Required: Bearer Token (JWT)

Responses

Status 200 : Successful Response

```
{
  "id" (Required): integer,
  "email" (Required): string,
  "full_name" (Required): string,
  "role" (Required): string,
  "phone": any,
  "is_active" (Required): boolean,
  "created_at" (Required): string,
}
```

Get Dashboard

GET /api/v1/admin/dashboard

Get real-time dashboard statistics.

Authorization Required: Bearer Token (JWT)

Responses

Status 200 : Successful Response

```
{
  "total_drivers": integer,
  "active_drivers": integer,
  "total_vehicles": integer,
  "active_vehicles": integer,
  "vehicles_in_maintenance": integer,
  "total_routes": integer,
  "active_routes": integer,
  "total_trips_today": integer,
  "completed_trips_today": integer,
  "total_revenue_today": number,
  "total_passengers_today": integer,
  "avg_crowding_score": number,
  "pending_maintenance": integer,
}
```

List Users

GET /api/v1/admin/users

List all users with optional filters.

Authorization Required: Bearer Token (JWT)

Parameters

Name	In	Type	Required	Description
role	query	string	No	
is_active	query	string	No	

Responses

Status 200 : Successful Response

```
[
  {
    "id" (Required): integer,
    "email" (Required): string,
    "full_name" (Required): string,
    "role" (Required): string,
    "phone": any,
    "is_active" (Required): boolean,
    "created_at" (Required): string,
  }
]
```

Status 422 : Validation Error

```
{
  "detail": [ {
    "loc" (Required): [ any ],
    "msg" (Required): string,
    "type" (Required): string,
    "input": any,
    "ctx": object,
  } ],
}
```

Create User

POST /api/v1/admin/users

Create a new user (admin only).

Authorization Required: Bearer Token (JWT)

Request Body

```
{
  "email" (Required): string,
  "password" (Required): string,
  "full_name" (Required): string,
  "role" (Required): string,
  "phone": any,
}
```

Responses

Status 201 : Successful Response

```
{
  "id" (Required): integer,
  "email" (Required): string,
  "full_name" (Required): string,
  "role" (Required): string,
  "phone": any,
  "is_active" (Required): boolean,
  "created_at" (Required): string,
}
```

Status 422 : Validation Error

```
{
  "detail": [ {
    "loc" (Required): [ any ],
    "msg" (Required): string,
    "type" (Required): string,
    "input": any,
    "ctx": object,
  } ],
}
```

Update User

PUT /api/v1/admin/users/{user_id}

Update a user's profile.

Authorization Required: Bearer Token (JWT)

Parameters

Name	In	Type	Required	Description
user_id	path	integer	Yes	

Request Body

```
{
  "full_name": any,
  "phone": any,
  "role": any,
  "is_active": any,
}
```

Responses

Status 200 : Successful Response

```
{
  "id" (Required): integer,
  "email" (Required): string,
  "full_name" (Required): string,
  "role" (Required): string,
  "phone": any,
  "is_active" (Required): boolean,
  "created_at" (Required): string,
}
```

Status 422 : Validation Error

```
{
  "detail": [ {
    "loc" (Required): [ any ],
    "msg" (Required): string,
    "type" (Required): string,
    "input": any,
    "ctx": object,
  } ],
}
```

Deactivate User

DELETE /api/v1/admin/users/{user_id}

Deactivate a user (soft delete).

Authorization Required: Bearer Token (JWT)

Parameters

Name	In	Type	Required	Description
user_id	path	integer	Yes	

Responses

Status 200 : Successful Response

```
{
  "message" (Required): string,
  "detail": any,
}
```

Status 422 : Validation Error

```
{
  "detail": [ {
    "loc" (Required): [ any ],
    "msg" (Required): string,
    "type" (Required): string,
    "input": any,
    "ctx": object,
  } ],
}
```

```
} ],  
}
```

List Drivers

GET /api/v1/admin/drivers

List all drivers.

Authorization Required: Bearer Token (JWT)

Parameters

Name	In	Type	Required	Description
status	query	string	No	

Responses

Status 200 : Successful Response

```
[  
  {  
    "id" (Required): integer,  
    "user_id" (Required): integer,  
    "license_number" (Required): string,  
    "license_expiry": any,  
    "garage_id": any,  
    "status" (Required): string,  
    "current_vehicle_id": any,  
    "current_route_id": any,  
    "total_trips_today": integer,  
    "total_trips_all_time": integer,  
    "rating": number,  
    "break_time_remaining": number,  
    "total_break_time_today": number,  
    "trips_since_last_break": integer,  
    "current_break_number": integer,  
    "current_shift": any,  
    "created_at" (Required): string,  
    "user": any,  
  }  
]
```

Status 422 : Validation Error

```
{  
  "detail": [ {  
    "loc" (Required): [ any ],  
    "msg" (Required): string,  
    "type" (Required): string,  
    "input": any,  
    "ctx": object,  
  } ],  
}
```


Create Driver

POST /api/v1/admin/drivers

Create a driver profile for an existing user.

Authorization Required: Bearer Token (JWT)

Request Body

```
{
  "user_id" (Required): integer,
  "license_number" (Required): string,
  "license_expiry": any,
  "garage_id": any,
}
```

Responses

Status 201 : Successful Response

```
{
  "id" (Required): integer,
  "user_id" (Required): integer,
  "license_number" (Required): string,
  "license_expiry": any,
  "garage_id": any,
  "status" (Required): string,
  "current_vehicle_id": any,
  "current_route_id": any,
  "total_trips_today": integer,
  "total_trips_all_time": integer,
  "rating": number,
  "break_time_remaining": number,
  "total_break_time_today": number,
  "trips_since_last_break": integer,
  "current_break_number": integer,
  "current_shift": any,
  "created_at" (Required): string,
  "user": any,
}
```

Status 422 : Validation Error

```
{
  "detail": [ {
    "loc" (Required): [ any ],
    "msg" (Required): string,
    "type" (Required): string,
    "input": any,
    "ctx": object,
  } ],
}
```

Update Driver

PUT /api/v1/admin/drivers/{driver_id}

Update driver fields.

Authorization Required: Bearer Token (JWT)

Parameters

Name	In	Type	Required	Description
driver_id	path	integer	Yes	

Request Body

```
{
  "license_number": any,
  "license_expiry": any,
  "status": any,
  "garage_id": any,
  "current_vehicle_id": any,
  "current_route_id": any,
}
```

Responses

Status 200 : Successful Response

```
{
  "id" (Required): integer,
  "user_id" (Required): integer,
  "license_number" (Required): string,
  "license_expiry": any,
  "garage_id": any,
  "status" (Required): string,
  "current_vehicle_id": any,
  "current_route_id": any,
  "total_trips_today": integer,
  "total_trips_all_time": integer,
  "rating": number,
  "break_time_remaining": number,
  "total_break_time_today": number,
  "trips_since_last_break": integer,
  "current_break_number": integer,
  "current_shift": any,
  "created_at" (Required): string,
  "user": any,
}
```

Status 422 : Validation Error

```
{
  "detail": [ {
    "loc" (Required): [ any ],
    "msg" (Required): string,
    "type" (Required): string,
    "input": any,
    "ctx": object,
  } ],
}
```

List Vehicles

GET /api/v1/admin/vehicles

List all vehicles.

Authorization Required: Bearer Token (JWT)

Parameters

Name	In	Type	Required	Description
status	query	string	No	

Responses

Status 200 : Successful Response

```
[
  {
    "id" (Required): integer,
    "plate_number" (Required): string,
    "model" (Required): string,
    "year": any,
    "capacity" (Required): integer,
    "status" (Required): string,
    "garage_id": any,
    "current_latitude": any,
    "current_longitude": any,
    "mileage": number,
    "fuel_level": number,
    "last_maintenance_date": any,
    "created_at" (Required): string,
  }
]
```

Status 422 : Validation Error

```
{
  "detail": [ {
    "loc" (Required): [ any ],
    "msg" (Required): string,
    "type" (Required): string,
    "input": any,
    "ctx": object,
  } ],
}
```

Create Vehicle

POST /api/v1/admin/vehicles

Add a new vehicle.

Authorization Required: Bearer Token (JWT)

Request Body

```
{
  "plate_number" (Required): string,
  "model" (Required): string,
  "year": any,
  "capacity": integer,
  "garage_id": any,
}
```

Responses

Status 201 : Successful Response

```
{
  "id" (Required): integer,
  "plate_number" (Required): string,
  "model" (Required): string,
  "year": any,
  "capacity" (Required): integer,
  "status" (Required): string,
  "garage_id": any,
  "current_latitude": any,
  "current_longitude": any,
  "mileage": number,
  "fuel_level": number,
  "last_maintenance_date": any,
  "created_at" (Required): string,
}
```

Status 422 : Validation Error

```
{
  "detail": [ {
    "loc" (Required): [ any ],
    "msg" (Required): string,
    "type" (Required): string,
    "input": any,
    "ctx": object,
  } ],
}
```

Update Vehicle

PUT /api/v1/admin/vehicles/{vehicle_id}

Update vehicle fields.

Authorization Required: Bearer Token (JWT)

Parameters

Name	In	Type	Required	Description
vehicle_id	path	integer	Yes	

Request Body

```
{
  "model": any,
  "capacity": any,
  "status": any,
  "garage_id": any,
  "current_latitude": any,
  "current_longitude": any,
  "mileage": any,
  "fuel_level": any,
}
```

Responses

Status **200** : Successful Response

```
{
  "id" (Required): integer,
  "plate_number" (Required): string,
  "model" (Required): string,
  "year": any,
  "capacity" (Required): integer,
  "status" (Required): string,
  "garage_id": any,
  "current_latitude": any,
  "current_longitude": any,
  "mileage": number,
  "fuel_level": number,
  "last_maintenance_date": any,
  "created_at" (Required): string,
}
```

Status **422** : Validation Error

```
{
  "detail": [ {
    "loc" (Required): [ any ],
    "msg" (Required): string,
    "type" (Required): string,
    "input": any,
    "ctx": object,
  } ],
}
```

List Routes

GET /api/v1/admin/routes

List all routes with their stops.

Authorization Required: Bearer Token (JWT)

Parameters

Name	In	Type	Required	Description
active_only	query	boolean	No	

Responses

Status 200 : Successful Response

```
[
  {
    "id" (Required): integer,
    "name" (Required): string,
    "start_location" (Required): string,
    "end_location" (Required): string,
    "distance_km": any,
    "estimated_time_minutes" (Required): number,
    "fare" (Required): number,
    "turnaround_time_minutes" (Required): number,
    "is_active" (Required): boolean,
    "stops": [ {
      "id" (Required): integer,
      "stop_name" (Required): string,
      "sequence_order" (Required): integer,
      "latitude": any,
      "longitude": any,
      "dwell_time_minutes" (Required): number,
    } ],
    "created_at" (Required): string,
  }
]
```

Status 422 : Validation Error

```
{
  "detail": [ {
    "loc" (Required): [ any ],
    "msg" (Required): string,
    "type" (Required): string,
    "input": any,
    "ctx": object,
  } ],
}
```

Create Route

POST /api/v1/admin/routes

Create a new route with optional stops.

Authorization Required: Bearer Token (JWT)

Request Body

```
{
  "name" (Required): string,
```

```
"start_location" (Required): string,
"end_location" (Required): string,
"distance_km": any,
"estimated_time_minutes" (Required): number,
"fare": number,
"turnaround_time_minutes": number,
"stops": any,
}
```

Responses

Status **201** : Successful Response

```
{
  "id" (Required): integer,
  "name" (Required): string,
  "start_location" (Required): string,
  "end_location" (Required): string,
  "distance_km": any,
  "estimated_time_minutes" (Required): number,
  "fare" (Required): number,
  "turnaround_time_minutes" (Required): number,
  "is_active" (Required): boolean,
  "stops": [ {
    "id" (Required): integer,
    "stop_name" (Required): string,
    "sequence_order" (Required): integer,
    "latitude": any,
    "longitude": any,
    "dwell_time_minutes" (Required): number,
  } ],
  "created_at" (Required): string,
}
```

Status **422** : Validation Error

```
{
  "detail": [ {
    "loc" (Required): [ any ],
    "msg" (Required): string,
    "type" (Required): string,
    "input": any,
    "ctx": object,
  } ],
}
```

List Assignments

GET /api/v1/admin/assignments

List rotation assignments with optional date/route filters.

Authorization Required: Bearer Token (JWT)

Parameters

Name	In	Type	Required	Description
shift_date	query	string	No	
route_id	query	string	No	

Responses

Status 200 : Successful Response

```
[
  {
    "id" (Required): integer,
    "route_id" (Required): integer,
    "driver_id" (Required): integer,
    "vehicle_id" (Required): integer,
    "shift_type" (Required): string,
    "position" (Required): string,
    "shift_date" (Required): string,
    "shift_start_time" (Required): string,
    "shift_end_time" (Required): string,
    "is_active" (Required): boolean,
    "created_at" (Required): string,
  }
]
```

Status 422 : Validation Error

```
{
  "detail": [ {
    "loc" (Required): [ any ],
    "msg" (Required): string,
    "type" (Required): string,
    "input": any,
    "ctx": object,
  } ],
}
```

Create Assignment

POST /api/v1/admin/assignments

Create a rotation assignment (admin override).

Authorization Required: Bearer Token (JWT)

Request Body

```
{
  "route_id" (Required): integer,
  "driver_id" (Required): integer,
  "vehicle_id" (Required): integer,
  "shift_type" (Required): string,
  "position" (Required): string,
  "shift_date" (Required): string,
```



```
"shift_start_time" (Required): string,
"shift_end_time" (Required): string,
}
```

Responses

Status **201** : Successful Response

```
{
  "id" (Required): integer,
  "route_id" (Required): integer,
  "driver_id" (Required): integer,
  "vehicle_id" (Required): integer,
  "shift_type" (Required): string,
  "position" (Required): string,
  "shift_date" (Required): string,
  "shift_start_time" (Required): string,
  "shift_end_time" (Required): string,
  "is_active" (Required): boolean,
  "created_at" (Required): string,
}
```

Status **422** : Validation Error

```
{
  "detail": [ {
    "loc" (Required): [ any ],
    "msg" (Required): string,
    "type" (Required): string,
    "input": any,
    "ctx": object,
  } ],
}
```

List Maintenance

GET /api/v1/admin/maintenance

List maintenance requests.

Authorization Required: Bearer Token (JWT)

Parameters

Name	In	Type	Required	Description
status	query	string	No	
vehicle_id	query	string	No	

Responses

Status **200** : Successful Response

```
[
  {
    "id" (Required): integer,
    "vehicle_id" (Required): integer,
    "requested_by_id" (Required): integer,
    "approved_by_id": any,
    "type" (Required): string,
    "status" (Required): string,
    "priority" (Required): integer,
    "title" (Required): string,
    "description": any,
    "estimated_cost": any,
    "actual_cost": any,
    "scheduled_date": any,
    "completed_date": any,
    "rejection_reason": any,
    "created_at" (Required): string,
  }
]
```

Status **422** : Validation Error

```
{
  "detail": [ {
    "loc" (Required): [ any ],
    "msg" (Required): string,
    "type" (Required): string,
    "input": any,
    "ctx": object,
  } ],
}
```

Approve Maintenance

POST /api/v1/admin/maintenance/{request_id}/approve

Approve a maintenance request.

Authorization Required: Bearer Token (JWT)

Parameters

Name	In	Type	Required	Description
request_id	path	integer	Yes	

Responses

Status **200** : Successful Response

```
{
  "id" (Required): integer,
  "vehicle_id" (Required): integer,
  "requested_by_id" (Required): integer,
  "approved_by_id": any,
  "type" (Required): string,
  "status" (Required): string,
```

```
"priority" (Required): integer,
"title" (Required): string,
"description": any,
"estimated_cost": any,
"actual_cost": any,
"scheduled_date": any,
"completed_date": any,
"rejection_reason": any,
"created_at" (Required): string,
}
```

Status 422 : Validation Error

```
{
  "detail": [ {
    "loc" (Required): [ any ],
    "msg" (Required): string,
    "type" (Required): string,
    "input": any,
    "ctx": object,
  } ],
}
```

Reject Maintenance

POST /api/v1/admin/maintenance/{request_id}/reject

Reject a maintenance request.

Authorization Required: Bearer Token (JWT)

Parameters

Name	In	Type	Required	Description
request_id	path	integer	Yes	
reason	query	string	Yes	Rejection reason

Responses

Status 200 : Successful Response

```
{
  "id" (Required): integer,
  "vehicle_id" (Required): integer,
  "requested_by_id" (Required): integer,
  "approved_by_id": any,
  "type" (Required): string,
  "status" (Required): string,
  "priority" (Required): integer,
  "title" (Required): string,
  "description": any,
  "estimated_cost": any,
  "actual_cost": any,
  "scheduled_date": any,
  "completed_date": any,
  "rejection_reason": any,
}
```

```
"created_at" (Required): string,
}
```

Status 422 : Validation Error

```
{
  "detail": [ {
    "loc" (Required): [ any ],
    "msg" (Required): string,
    "type" (Required): string,
    "input": any,
    "ctx": object,
  } ],
}
```

Complete Maintenance

POST /api/v1/admin/maintenance/{request_id}/complete

Complete a maintenance request.

Authorization Required: Bearer Token (JWT)

Parameters

Name	In	Type	Required	Description
request_id	path	integer	Yes	
actual_cost	query	string	No	

Responses

Status 200 : Successful Response

```
{
  "id" (Required): integer,
  "vehicle_id" (Required): integer,
  "requested_by_id" (Required): integer,
  "approved_by_id": any,
  "type" (Required): string,
  "status" (Required): string,
  "priority" (Required): integer,
  "title" (Required): string,
  "description": any,
  "estimated_cost": any,
  "actual_cost": any,
  "scheduled_date": any,
  "completed_date": any,
  "rejection_reason": any,
  "created_at" (Required): string,
}
```

Status 422 : Validation Error

```
{
  "detail": [ {
    "loc" (Required): [ any ],
    "msg" (Required): string,
    "type" (Required): string,
    "input": any,
    "ctx": object,
  } ],
}
```

Get Daily Report

GET /api/v1/admin/reports/daily

Get or generate a daily report.

Authorization Required: Bearer Token (JWT)

Parameters

Name	In	Type	Required	Description
report_date	query	string	No	

Responses

Status 200 : Successful Response

```
{
  "id" (Required): integer,
  "report_date" (Required): string,
  "total_trips" (Required): integer,
  "completed_trips" (Required): integer,
  "cancelled_trips" (Required): integer,
  "total_revenue" (Required): number,
  "total_passengers" (Required): integer,
  "avg_crowding_score" (Required): number,
  "on_time_percentage" (Required): number,
  "total_maintenance_requests" (Required): integer,
  "active_vehicles" (Required): integer,
  "active_drivers" (Required): integer,
  "extra_dispatches" (Required): integer,
  "created_at" (Required): string,
}
```

Status 422 : Validation Error

```
{
  "detail": [ {
    "loc" (Required): [ any ],
    "msg" (Required): string,
    "type" (Required): string,
    "input": any,
    "ctx": object,
  } ],
}
```

Get Reports Range

GET /api/v1/admin/reports/range

Get daily reports for a date range.

Authorization Required: Bearer Token (JWT)

Parameters

Name	In	Type	Required	Description
start_date	query	string	Yes	
end_date	query	string	Yes	

Responses

Status 200 : Successful Response

```
[
  {
    "id" (Required): integer,
    "report_date" (Required): string,
    "total_trips" (Required): integer,
    "completed_trips" (Required): integer,
    "cancelled_trips" (Required): integer,
    "total_revenue" (Required): number,
    "total_passengers" (Required): integer,
    "avg_crowding_score" (Required): number,
    "on_time_percentage" (Required): number,
    "total_maintenance_requests" (Required): integer,
    "active_vehicles" (Required): integer,
    "active_drivers" (Required): integer,
    "extra_dispatches" (Required): integer,
    "created_at" (Required): string,
  }
]
```

Status 422 : Validation Error

```
{
  "detail": [ {
    "loc" (Required): [ any ],
    "msg" (Required): string,
    "type" (Required): string,
    "input": any,
    "ctx": object,
  } ],
}
```

List Audit Logs

GET /api/v1/admin/audit-logs

List recent audit log entries.

Authorization Required: Bearer Token (JWT)

Parameters

Name	In	Type	Required	Description
limit	query	integer	No	
entity_type	query	string	No	

Responses

Status 200 : Successful Response

```
[
  {
    "id" (Required): integer,
    "user_id": any,
    "action" (Required): string,
    "entity_type" (Required): string,
    "entity_id": any,
    "old_values": any,
    "new_values": any,
    "ip_address": any,
    "created_at" (Required): string,
  }
]
```

Status 422 : Validation Error

```
{
  "detail": [ {
    "loc" (Required): [ any ],
    "msg" (Required): string,
    "type" (Required): string,
    "input": any,
    "ctx": object,
  } ],
}
```

Get Garage

GET /api/v1/admin/garage

Get garage information.

Authorization Required: Bearer Token (JWT)

Responses

Status 200 : Successful Response

```
{
  "id" (Required): integer,
  "name" (Required): string,
  "address": any,
  "latitude": any,
  "longitude": any,
  "total_capacity" (Required): integer,
  "current_occupancy" (Required): integer,
  "created_at" (Required): string,
}
```

Get Dashboard

GET /api/v1/manager/dashboard

Get dashboard statistics.

Authorization Required: Bearer Token (JWT)

Responses

Status 200 : Successful Response

```
{
  "total_drivers": integer,
  "active_drivers": integer,
  "total_vehicles": integer,
  "active_vehicles": integer,
  "vehicles_in_maintenance": integer,
  "total_routes": integer,
  "active_routes": integer,
  "total_trips_today": integer,
  "completed_trips_today": integer,
  "total_revenue_today": number,
  "total_passengers_today": integer,
  "avg_crowding_score": number,
  "pending_maintenance": integer,
}
```

List Vehicles

GET /api/v1/manager/vehicles

View all vehicles and their current status.

Authorization Required: Bearer Token (JWT)

Responses

Status 200 : Successful Response

```
[
  {
    "id" (Required): integer,
```



```
"plate_number" (Required): string,
"model" (Required): string,
"year": any,
"capacity" (Required): integer,
"status" (Required): string,
"garage_id": any,
"current_latitude": any,
"current_longitude": any,
"mileage": number,
"fuel_level": number,
"last_maintenance_date": any,
"created_at" (Required): string,
}
]
```

List Drivers

GET /api/v1/manager/drivers

View all drivers and their current status.

Authorization Required: Bearer Token (JWT)

Responses

Status 200 : Successful Response

```
[
  {
    "id" (Required): integer,
    "user_id" (Required): integer,
    "license_number" (Required): string,
    "license_expiry": any,
    "garage_id": any,
    "status" (Required): string,
    "current_vehicle_id": any,
    "current_route_id": any,
    "total_trips_today": integer,
    "total_trips_all_time": integer,
    "rating": number,
    "break_time_remaining": number,
    "total_break_time_today": number,
    "trips_since_last_break": integer,
    "current_break_number": integer,
    "current_shift": any,
    "created_at" (Required): string,
    "user": any,
  }
]
```

List Routes

GET /api/v1/manager/routes

View all routes.

Authorization Required: Bearer Token (JWT)

Responses

Status **200** : Successful Response

```
[
  {
    "id" (Required): integer,
    "name" (Required): string,
    "start_location" (Required): string,
    "end_location" (Required): string,
    "distance_km": any,
    "estimated_time_minutes" (Required): number,
    "fare" (Required): number,
    "turnaround_time_minutes" (Required): number,
    "is_active" (Required): boolean,
    "stops": [ {
      "id" (Required): integer,
      "stop_name" (Required): string,
      "sequence_order" (Required): integer,
      "latitude": any,
      "longitude": any,
      "dwell_time_minutes" (Required): number,
    } ],
    "created_at" (Required): string,
  }
]
```

List Trips

GET /api/v1/manager/trips

View trips with optional filters.

Authorization Required: Bearer Token (JWT)

Parameters

Name	In	Type	Required	Description
trip_date	query	string	No	
route_id	query	string	No	
status	query	string	No	

Responses

Status **200** : Successful Response

```
[
  {
    "id" (Required): integer,
    "driver_id" (Required): integer,
    "vehicle_id" (Required): integer,
    "route_id" (Required): integer,
    "rotation_assignment_id": any,
    "direction" (Required): string,
    "status" (Required): string,
    "trip_number": any,
```

```
"scheduled_start" (Required): string,
"scheduled_end": any,
"actual_start": any,
"actual_end": any,
"passenger_count": integer,
"crowding_score": number,
"is_crowded": boolean,
"driver_crowding_report": boolean,
"fare_collected": number,
"is_extra_dispatch": boolean,
"is_late": boolean,
"notes": any,
"created_at" (Required): string,
}
]
```

Status 422 : Validation Error

```
{
  "detail": [ {
    "loc" (Required): [ any ],
    "msg" (Required): string,
    "type" (Required): string,
    "input": any,
    "ctx": object,
  } ],
}
```

List Assignments

GET /api/v1/manager/assignments

View rotation assignments.

Authorization Required: Bearer Token (JWT)

Parameters

Name	In	Type	Required	Description
shift_date	query	string	No	

Responses

Status 200 : Successful Response

```
[
  {
    "id" (Required): integer,
    "route_id" (Required): integer,
    "driver_id" (Required): integer,
    "vehicle_id" (Required): integer,
    "shift_type" (Required): string,
    "position" (Required): string,
    "shift_date" (Required): string,
    "shift_start_time" (Required): string,
    "shift_end_time" (Required): string,
    "is_active" (Required): boolean,
```

```
    "created_at" (Required): string,
  }
]
```

Status 422 : Validation Error

```
{
  "detail": [ {
    "loc" (Required): [ any ],
    "msg" (Required): string,
    "type" (Required): string,
    "input": any,
    "ctx": object,
  } ],
}
```

List Maintenance

GET /api/v1/manager/maintenance

View maintenance requests.

Authorization Required: Bearer Token (JWT)

Parameters

Name	In	Type	Required	Description
status	query	string	No	

Responses

Status 200 : Successful Response

```
[
  {
    "id" (Required): integer,
    "vehicle_id" (Required): integer,
    "requested_by_id" (Required): integer,
    "approved_by_id": any,
    "type" (Required): string,
    "status" (Required): string,
    "priority" (Required): integer,
    "title" (Required): string,
    "description": any,
    "estimated_cost": any,
    "actual_cost": any,
    "scheduled_date": any,
    "completed_date": any,
    "rejection_reason": any,
    "created_at" (Required): string,
  }
]
```

Status 422 : Validation Error

```
{
  "detail": [ {
    "loc" (Required): [ any ],
    "msg" (Required): string,
    "type" (Required): string,
    "input": any,
    "ctx": object,
  } ],
}
```

Create Maintenance Request

POST /api/v1/manager/maintenance

Submit a new maintenance request.

Authorization Required: Bearer Token (JWT)

Request Body

```
{
  "vehicle_id" (Required): integer,
  "type": string,
  "title" (Required): string,
  "description": any,
  "priority": integer,
  "estimated_cost": any,
  "scheduled_date": any,
}
```

Responses

Status 201 : Successful Response

```
{
  "id" (Required): integer,
  "vehicle_id" (Required): integer,
  "requested_by_id" (Required): integer,
  "approved_by_id": any,
  "type" (Required): string,
  "status" (Required): string,
  "priority" (Required): integer,
  "title" (Required): string,
  "description": any,
  "estimated_cost": any,
  "actual_cost": any,
  "scheduled_date": any,
  "completed_date": any,
  "rejection_reason": any,
  "created_at" (Required): string,
}
```

Status 422 : Validation Error

```
{
  "detail": [ {
```

```
"loc" (Required): [ any ],
"msg" (Required): string,
"type" (Required): string,
"input": any,
"context": object,
} ],
}
```

Approve Maintenance

POST /api/v1/manager/maintenance/{request_id}/approve

Approve a pending maintenance request.

Authorization Required: Bearer Token (JWT)

Parameters

Name	In	Type	Required	Description
request_id	path	integer	Yes	

Responses

Status 200 : Successful Response

```
{
  "id" (Required): integer,
  "vehicle_id" (Required): integer,
  "requested_by_id" (Required): integer,
  "approved_by_id": any,
  "type" (Required): string,
  "status" (Required): string,
  "priority" (Required): integer,
  "title" (Required): string,
  "description": any,
  "estimated_cost": any,
  "actual_cost": any,
  "scheduled_date": any,
  "completed_date": any,
  "rejection_reason": any,
  "created_at" (Required): string,
}
```

Status 422 : Validation Error

```
{
  "detail": [ {
    "loc" (Required): [ any ],
    "msg" (Required): string,
    "type" (Required): string,
    "input": any,
    "context": object,
  } ],
}
```

Get Crowded Routes

GET /api/v1/manager/crowding

Get currently crowded routes.

Authorization Required: Bearer Token (JWT)

Responses

Status 200 : Successful Response

Get Route Crowding

GET /api/v1/manager/crowding/{route_id}

Get crowding statistics for a specific route.

Authorization Required: Bearer Token (JWT)

Parameters

Name	In	Type	Required	Description
route_id	path	integer	Yes	

Responses

Status 200 : Successful Response

Status 422 : Validation Error

```
{
  "detail": [ {
    "loc" (Required): [ any ],
    "msg" (Required): string,
    "type" (Required): string,
    "input": any,
    "ctx": object,
  } ],
}
```

Get My Profile

GET /api/v1/drivers/me

Get the current driver's profile.

Authorization Required: Bearer Token (JWT)

Responses

Status 200 : Successful Response

```
{
  "id" (Required): integer,
  "user_id" (Required): integer,
  "license_number" (Required): string,
  "license_expiry": any,
  "garage_id": any,
  "status" (Required): string,
  "current_vehicle_id": any,
  "current_route_id": any,
  "total_trips_today": integer,
  "total_trips_all_time": integer,
  "rating": number,
  "break_time_remaining": number,
  "total_break_time_today": number,
  "trips_since_last_break": integer,
  "current_break_number": integer,
  "current_shift": any,
  "created_at" (Required): string,
  "user": any,
}
```

Get My Status

GET /api/v1/drivers/me/status

Get detailed driver status including shift, break, and next trip info.

Authorization Required: Bearer Token (JWT)

Responses

Status 200 : Successful Response

Get My Trips

GET /api/v1/drivers/me/trips

Get driver's trips, optionally filtered by status.

Authorization Required: Bearer Token (JWT)

Parameters

Name	In	Type	Required	Description
status	query	string	No	

Responses

Status 200 : Successful Response

```
[
  {
    "id" (Required): integer,
```



```
"driver_id" (Required): integer,
"vehicle_id" (Required): integer,
"route_id" (Required): integer,
"rotation_assignment_id": any,
"direction" (Required): string,
"status" (Required): string,
"trip_number": any,
"scheduled_start" (Required): string,
"scheduled_end": any,
"actual_start": any,
"actual_end": any,
"passenger_count": integer,
"crowding_score": number,
"is_crowded": boolean,
"driver_crowding_report": boolean,
"fare_collected": number,
"is_extra_dispatch": boolean,
"is_late": boolean,
"notes": any,
"created_at" (Required): string,
}
]
```

Status 422 : Validation Error

```
{
  "detail": [ {
    "loc" (Required): [ any ],
    "msg" (Required): string,
    "type" (Required): string,
    "input": any,
    "ctx": object,
  } ],
}
```

Start Trip

POST /api/v1/drivers/me/trips/{trip_id}/start

Start a scheduled trip.

Authorization Required: Bearer Token (JWT)

Parameters

Name	In	Type	Required	Description
trip_id	path	integer	Yes	

Responses

Status 200 : Successful Response

```
{
  "id" (Required): integer,
  "driver_id" (Required): integer,
  "vehicle_id" (Required): integer,
  "route_id" (Required): integer,
```

```
"rotation_assignment_id": any,
"direction" (Required): string,
"status" (Required): string,
"trip_number": any,
"scheduled_start" (Required): string,
"scheduled_end": any,
"actual_start": any,
"actual_end": any,
"passenger_count": integer,
"crowding_score": number,
"is_crowded": boolean,
"driver_crowding_report": boolean,
"fare_collected": number,
"is_extra_dispatch": boolean,
"is_late": boolean,
"notes": any,
"created_at" (Required): string,
}
```

Status 422 : Validation Error

```
{
  "detail": [ {
    "loc" (Required): [ any ],
    "msg" (Required): string,
    "type" (Required): string,
    "input": any,
    "ctx": object,
  } ],
}
```

Complete Trip

POST /api/v1/drivers/me/trips/{trip_id}/complete

Complete an active trip.

Authorization Required: Bearer Token (JWT)

Parameters

Name	In	Type	Required	Description
trip_id	path	integer	Yes	
passenger_count	query	string	No	
fare_collected	query	string	No	

Responses

Status 200 : Successful Response

```
{
  "id" (Required): integer,
  "driver_id" (Required): integer,
  "vehicle_id" (Required): integer,
  "route_id" (Required): integer,
```

```
"rotation_assignment_id": any,
"direction" (Required): string,
"status" (Required): string,
"trip_number": any,
"scheduled_start" (Required): string,
"scheduled_end": any,
"actual_start": any,
"actual_end": any,
"passenger_count": integer,
"crowding_score": number,
"is_crowded": boolean,
"driver_crowding_report": boolean,
"fare_collected": number,
"is_extra_dispatch": boolean,
"is_late": boolean,
"notes": any,
"created_at" (Required): string,
}
```

Status 422 : Validation Error

```
{
  "detail": [ {
    "loc" (Required): [ any ],
    "msg" (Required): string,
    "type" (Required): string,
    "input": any,
    "ctx": object,
  } ],
}
```

Report Crowding

POST /api/v1/drivers/me/trips/{trip_id}/crowding

Report crowding on an active trip.

Authorization Required: Bearer Token (JWT)

Parameters

Name	In	Type	Required	Description
trip_id	path	integer	Yes	
passenger_count	query	integer	Yes	

Responses

Status 200 : Successful Response

Status 422 : Validation Error

```
{
  "detail": [ {
    "loc" (Required): [ any ],
    "msg" (Required): string,
    "type" (Required): string,
```

```
"input": any,  
"ctx": object,  
} ],  
}
```

Get Break Status

GET /api/v1/drivers/me/break/status

Check break eligibility and remaining time.

Authorization Required: Bearer Token (JWT)

Responses

Status 200 : Successful Response

Start Break

POST /api/v1/drivers/me/break/start

Request a break.

Authorization Required: Bearer Token (JWT)

Request Body

```
{  
  "requested_minutes" (Required): number,  
}
```

Responses

Status 200 : Successful Response

Status 422 : Validation Error

```
{  
  "detail": [ {  
    "loc" (Required): [ any ],  
    "msg" (Required): string,  
    "type" (Required): string,  
    "input": any,  
    "ctx": object,  
  } ],  
}
```

End Break

POST /api/v1/drivers/me/break/end

End current break and return to active duty.

Authorization Required: Bearer Token (JWT)

Responses

Status 200 : Successful Response

Submit Gps

POST /api/v1/drivers/me/gps

Submit a GPS data point.

Authorization Required: Bearer Token (JWT)

Request Body

```
{
  "vehicle_id" (Required): integer,
  "trip_id": any,
  "latitude" (Required): number,
  "longitude" (Required): number,
  "speed_kmh": any,
  "heading": any,
}
```

Responses

Status 201 : Successful Response

```
{
  "id" (Required): integer,
  "vehicle_id" (Required): integer,
  "trip_id": any,
  "latitude" (Required): number,
  "longitude" (Required): number,
  "speed_kmh": any,
  "heading": any,
  "recorded_at" (Required): string,
}
```

Status 422 : Validation Error

```
{
  "detail": [ {
    "loc" (Required): [ any ],
    "msg" (Required): string,
    "type" (Required): string,
    "input": any,
    "ctx": object,
  } ],
}
```

Get Notifications

GET /api/v1/drivers/me/notifications

Get driver's notifications.

Authorization Required: Bearer Token (JWT)

Parameters

Name	In	Type	Required	Description
unread_only	query	boolean	No	

Responses

Status 200 : Successful Response

```
[
  {
    "id" (Required): integer,
    "user_id" (Required): integer,
    "title" (Required): string,
    "message" (Required): string,
    "notification_type": any,
    "is_read" (Required): boolean,
    "created_at" (Required): string,
  }
]
```

Status 422 : Validation Error

```
{
  "detail": [ {
    "loc" (Required): [ any ],
    "msg" (Required): string,
    "type" (Required): string,
    "input": any,
    "ctx": object,
  } ],
}
```

Mark Notification Read

POST /api/v1/drivers/me/notifications/{notification_id}/read

Mark a notification as read.

Authorization Required: Bearer Token (JWT)

Parameters

Name	In	Type	Required	Description
notification_id	path	integer	Yes	

Responses

Status 200 : Successful Response

```
{
  "message" (Required): string,
  "detail": any,
}
```

Status 422 : Validation Error

```
{
  "detail": [ {
    "loc" (Required): [ any ],
    "msg" (Required): string,
    "type" (Required): string,
    "input": any,
    "ctx": object,
  } ],
}
```

Mark All Notifications Read

POST /api/v1/drivers/me/notifications/read-all

Mark all notifications as read.

Authorization Required: Bearer Token (JWT)

Responses

Status 200 : Successful Response

```
{
  "message" (Required): string,
  "detail": any,
}
```

Ingest Telemetry

POST /api/v1/hardware/iot

Ingest high-frequency engine telemetry and battery data.

Authorization Required: Bearer Token (JWT)

Request Body

```
{
  "vehicle_id" (Required): integer,
  "engine_temp_c" (Required): number,
  "oil_pressure_psi" (Required): number,
  "battery_voltage" (Required): number,
  "brake_pad_mm" (Required): number,
}
```

Responses

Status **200** : Successful Response

```
{
  "message" (Required): string,
  "detail": any,
}
```

Status **422** : Validation Error

```
{
  "detail": [ {
    "loc" (Required): [ any ],
    "msg" (Required): string,
    "type" (Required): string,
    "input": any,
    "ctx": object,
  } ],
}
```

Anpr Webhook

POST /api/v1/hardware/anpr

Gate Automation matching OCR plates.

Authorization Required: Bearer Token (JWT)

Request Body

```
{
  "camera_id" (Required): string,
  "plate_number" (Required): string,
}
```

Responses

Status **200** : Successful Response

```
{
  "message" (Required): string,
```



```
{
  "detail": any,
}
```

Status 422 : Validation Error

```
{
  "detail": [ {
    "loc" (Required): [ any ],
    "msg" (Required): string,
    "type" (Required): string,
    "input": any,
    "ctx": object,
  } ],
}
```

Process Camera Inference

POST /api/v1/hardware/camera

YOLOv8 inference webhook for crowding passenger counts.

Authorization Required: Bearer Token (JWT)

Request Body

```
{
  "camera_id" (Required): string,
  "detected_count" (Required): integer,
  "confidence_score" (Required): number,
}
```

Responses

Status 200 : Successful Response

```
{
  "message" (Required): string,
  "detail": any,
}
```

Status 422 : Validation Error

```
{
  "detail": [ {
    "loc" (Required): [ any ],
    "msg" (Required): string,
    "type" (Required): string,
    "input": any,
    "ctx": object,
  } ],
}
```

Validate Ticket

POST /api/v1/tickets/validate

NFC/QR scanning with row-level locking (`with_for_update()`) to prevent double-scan fraud.

Authorization Required: Bearer Token (JWT)

Request Body

```
{
  "ticket_code" (Required): string,
  "device_id": any,
}
```

Responses

Status 200 : Successful Response

```
{
  "valid" (Required): boolean,
  "status" (Required): string,
  "message" (Required): string,
}
```

Status 422 : Validation Error

```
{
  "detail": [ {
    "loc" (Required): [ any ],
    "msg" (Required): string,
    "type" (Required): string,
    "input": any,
    "ctx": object,
  } ],
}
```

Root

GET /

Health check — confirms the API is running.

Responses

Status 200 : Successful Response

Security & Rate Limiting

Authentication

The API strictly uses JWT tokens. Due to Python 3.13 constraints, password hashes are generated using raw `bcrypt`.

Rate Limiting (SlowAPI with Redis)

To prevent abuse across Uvicorn workers, rate limiting is handled by SlowAPI backed by Redis:

- **Unauthenticated (Login/Register):** 10/minute
- **Standard Driver API:** 100/minute
- **Hardware Webhooks (GPS & Engine):** 300/minute
- **Internal Admin Scrapes:** 600/minute

WebSockets

The Smart Bus Garage uses Redis-backed WebSockets managed by `ConnectionManager` to push real-time updates for telemetry, alerts, and operational events.

Connection Endpoint

WS `/api/v1/ws/dashboard/{role}`

Connect to this endpoint using the appropriate `UserRole` (`admin`, `manager`, `driver`). Authentication context must be provided. The API relies on Zustand state management to consume these priority envelopes payload.

Emitted Events (Priority Envelopes)

HARDWARE_ALERT (**CRITICAL** | **HIGH**)

Triggered by IoT Telemetry breaches (e.g., Temperature > 105C, Battery < 11.8V).

```
{
  "priority": "CRITICAL",
  "type": "HARDWARE_ALERT",
  "vehicle_id": 1,
  "message": "Engine temperature exceeded 105C"
}
```

GATE_AUTH_EVENT (**HIGH** | **CRITICAL**)

Triggered by ANPR Gate cameras upon vehicle arrival.

```
{
  "priority": "HIGH",
  "type": "GATE_AUTH_GRANTED",
  "plate_number": "XYZ-123",
  "message": "Authorized vehicle entered Main Gate"
}
```

CROWDING_DISPATCH (CRITICAL)

Triggered when YOLOv8 AI cameras detect >90% crowding and a new driver is auto-dispatched via the ping-pong algorithm.

```
{  
  "priority": "CRITICAL",  
  "type": "CROWDING_DISPATCH",  
  "route_id": 5,  
  "new_trip_id": 102,  
  "driver_id": 8  
}
```